

Cryptology → Assignment 5

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- 1) Encrypt the message "ENJOY YOUR JOURNEY" using Rail fence with depth 2

Depth 2

E N J O Y R O R N E Y
N O Y U T U N Y

Cipher: EJYD RORE NOYUTUNY.

Depth 3

E N J O Y R O R N E Y
J O U T U N Y

Cipher: EYRR NOYUTUNY JDOE

- 2) Encrypt the message as per your choice using Rail Fence with depth 2 & 3. Plaintext is GOOD DAY

Depth 2

G O D A Y
O D A Y

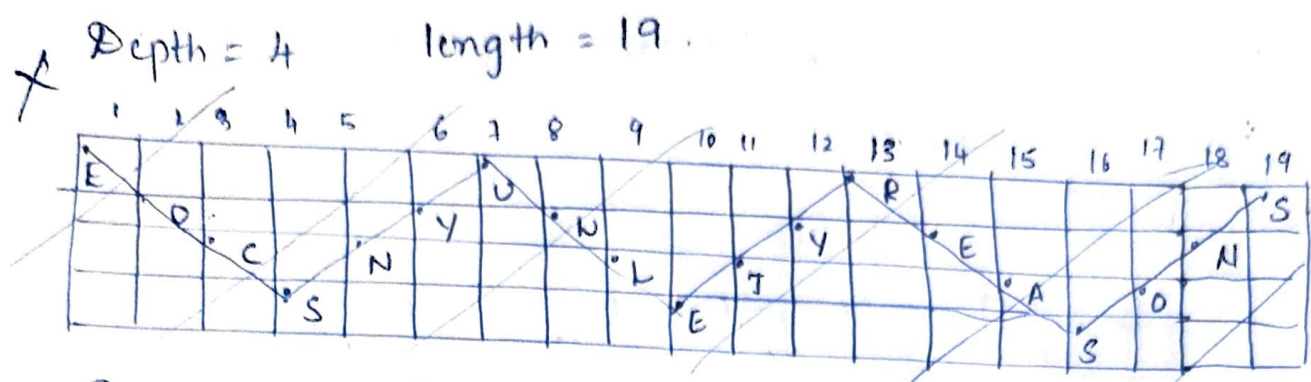
Cipher: GODYODA

Depth 3

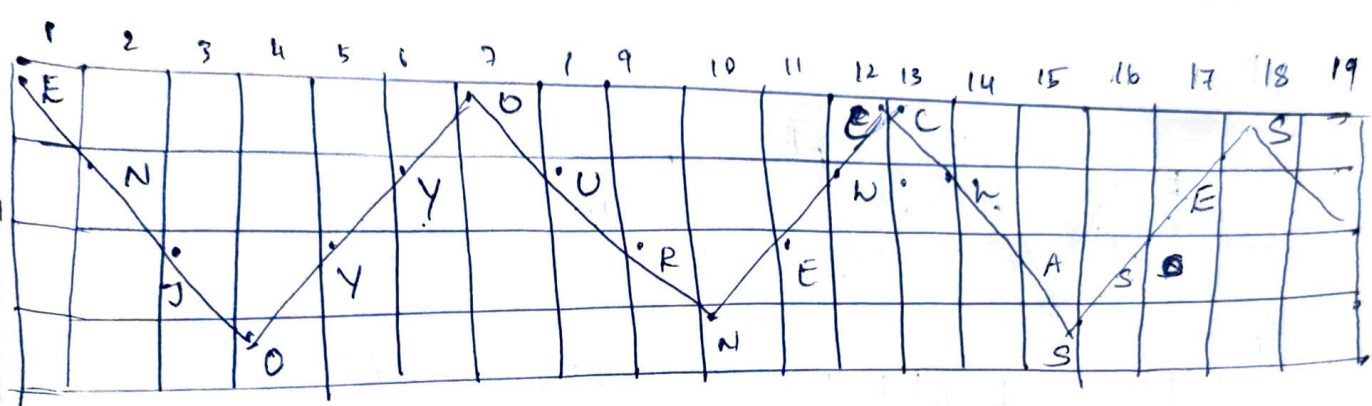
G O D A Y
O D A Y

Cipher: GDOADA Y

3. Decrypt the cipher "EOCSNYUNLET YREA
SONS", Encrypted using Rail fence with depth 4



Depth = 4 length = 19.



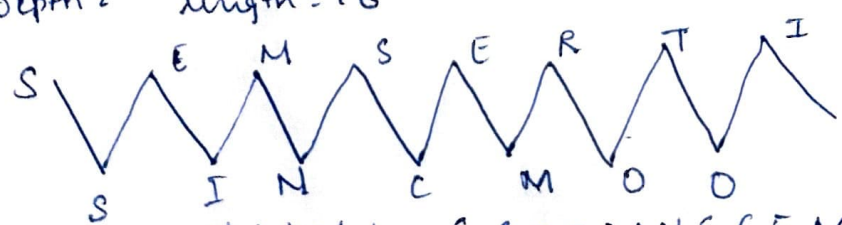
Plaintext :- ENJOY YOUR NEW CLASSES.

4) Decrypt the cipher "BOOI GUY UBOSNUUTYRKAS"¹², encrypted using Rail fence with depth 3.
depth = 3 length = 20.



Plaintext: BUY YOUR BOOKS IN AUGUST

5) Decrypt the cipher "SEMGERT I SINCMOO" encrypted using rail fence with depth 2.
Depth 2 length = 16



Plaintext: SSEIMNSCEMROTOJ

Permutation

1. Encrypt the following message "We are discovered" using the key 5 3 1 2 4.

5	3	1	2	4
W	E	A	R	E
D	I	S	C	O
V	E	R	E	D

Cipher text : A S R R C E E I E E O D N D V

DES

1. Using the initial permutation table, determine the output of the initial permutation box when the input is given in hexadecimal as : 0x002000000001

Binary form :

{	0000	0000	0000	0010
	0000	0000	0000	0000
	0000	0000	0000	0000
	0000	0000	0000	0000
	0000	0000	0000	0001

↓
input to box (plaintext)

0000 0000 0000 0010 0000 0000 0000 0000 0000
0000 0000 0000 000000 0010

Input bit 64 = 1 , o/p position 25 = 1 } By using P-Table.

Input bit 15 = 1 , o/p position 63 = 1

0	0	0	0	0	0	8	0	0	0	0	0
0000	0000	0000	0000	0000	0000	1000	0000	0000	0000	0000	0000
0000	0000	0000	0000	0010							

o/p in hexadecimal : [0x0000080000000002]

- 2) Given the 4 bit Plaintext $[P] 1101$ & Key $[K] 1011$
Calculate the resulting Ciphertext $[C]$ using a bitwise Exclusive-OR $[XOR]$ operation.

Solu

$$P = 1101$$

$$K = 1011$$

$$Ciphertext = 0110$$

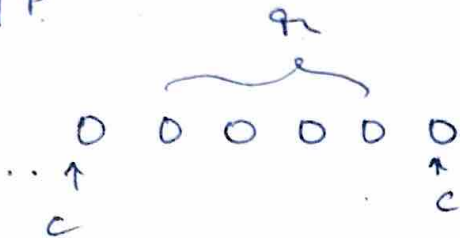
$$\therefore \boxed{Ciphertext = 0110}$$

XOR

Same bit = 0

diff bit = 1

3. The input to S-box 1 is 000000. what is O/P



so, 0^{th} row & 0^{th} column in the S-box is 14,

$$\boxed{O/P = 1110}$$

$$14 \rightarrow \begin{matrix} 2^3 & 2^2 & 2^1 & 2^0 \\ \hline 1 & 1 & 1 & 0 \end{matrix}$$